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Personalized Recommender Systems under Artificial Intelligence Breakthrough

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Abstract

This review critically examines the evolution of AI-enhanced recommender systems, with a focus on addressing the challenges of data sparsity and scalability. It presents various novel deep learning frameworks from published papers and industry leaders, like Google and Netflix, revealing a shift from traditional models to AI algorithms that capture more user behavior information. Key findings from leading platforms validate the effectiveness of these methods in large-scale applications. The review concludes with the potential of open-source initiatives to democratize AI development and suggests future research directions that emphasize ethical AI and advanced multimodal personalization.

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1 Introduction

In the current information and disinformation age, recommendations from tech platforms are influencing society profoundly. Companies compete for our attention to maximize app usage and advertisement revenue, treating human attention and behaviors as valuable products, and focusing on engagement, growth, and ad revenue. Surveillance capitalism profits from tracking users' daily data, impacting human futures on a large scale. Unethical techniques, like positive intermittent reinforcement and psychological manipulation, are used to keep users engaged. Social media experiments, like the Facebook emotional contagion experiment, can influence real-world behavior without users' awareness because technology has transitioned from a tool-based environment to a manipulative one, affecting our sense of perfection and affirmation. Artificial Intelligence (AI) can shape our mindset and psychology, as learnable algorithms could potentially create echo chambers, polarizing society, and false information, thereby destabilizing society, and even humans struggle to identify the real truth in this false-information world, not to mention the AI-biased problem. Therefore, this review delves into the design details of recommender systems (RS) in the ever-growing AI era, unraveling their complexities and identifying areas of innovation and challenge, and the pivotal role they play in user engagement and satisfaction from the case studies in the entertainment industry.

In this review, the historical evolution of RS is outlined, marking significant milestones and paradigm shifts. Starting with the traditional RS based on the book [1], this review explains Content-Based System (CB), which matches items to a user's preferences, to more sophisticated techniques like neighborhood-based Collaborative Filtering (CF) and Matrix Factorization methods (MF), which brought a deeper understanding of user-item interactions. Each of the methods is associated with its updated research work by Deldjoo et al.[2], He et al.[3], and Ning et al.[4] respectively. These essential systems have incorporated complex algorithms and deep learning networks due to progressive AI development, reflecting AI adaption in RS in various industries. However, they face criticisms regarding scalability and sensitivity to noisy and sparse data.

This review provides a roadmap for the evolution of RS, answering the following questions:

1. What are the innovations and challenges of AI in the evolution of more personalized recommendation systems?
2. How can advanced deep-learning neural networks in recommendation systems be designed to actually address scalability and data sparsity challenges?
3. What are the ethical considerations and business implications that arise in the implementation of AI-based recommendation systems in various industries?

Integrating deep learning into RS has faced its share of challenges, with criticisms of scalability and the models' sensitivity to the ubiquitous presence of noisy and sparse data within large real-world data with the exponential growth of users and their content. With the advent of more mature deep-learning technologies, different topologies of neural networks have been devised, each addressing different sophisticated fields of RS. Restricted Boltzmann Machine (RBM) laid the groundwork, Hinton et al.[5] shows the paralleling traditional matrix factorization methods in pattern recognition efficacy. Contemporary neural networks, like Auto-Encoder for Recommendations (AutoRec) by Sedhain et al.[6], have refined this approach, albeit remaining the issue of the inherent sparsity in user data. Recurrent Neural Networks (RNNs), particularly through the Gated Recurrent Unit for Recommendations (GRU4Rec) model, have brought

sequential data analysis to the progress of session-based recommendations by Hidasi et al.[7]. Neural Collaborative Filtering (NCF) by He et al.[8] has merged the strengths of Generalized Matrix Factorization models (GMFs) and Multi-Layer Perceptron (MLP), paving a hybrid path forward. Deep factorization machines (DeepFMs) by Guo et al.[9] have been further introduced, operating by feeding feature data into both a factorization machine and a deep neural network, which seeks to capture both low and high-order feature interactions. Meanwhile, Generative Adversarial Networks (GANs), adapted for recommendation systems (RecGANs), have showcased their efficacy in generating comprehensive user-item rating matrices to handle data sparsity indicated by Bharadhwaj et al.[10]. Collectively, these cutting-edge methodologies represent a great and progressive step in the field, underscoring the expansive potential of deep learning within the RS domain.

The impact of these solutions on the real world is far-reaching, influencing not only the technical landscape but also the entertainment industry's consumption patterns and business models. This review discusses the implications of these advancements and the ongoing research for more refined and user-centric RS. The scope of this review is further defined by the inclusion of detailed case studies from industry leaders like YouTube [11] and Netflix [1], exploring the complexity of these systems and their significance in an era of exponential AI growth, and briefly illustrate the real-world challenges they face.

This review will not delve into the detailed mathematics that underpin RS, the focus is on the state-of-the-art research progress and design in RS. The boundaries of the review are set to maintain clarity and relevance to the promising research results of RS. Also, knowledge of RS is not assumed, therefore, I will provide some essential background knowledge that is necessary to understand RS and the solutions that are proposed in the following section. We do take for granted though that the reader is familiar with some fundamental knowledge of Neural Networks (NNs), such as Convolutional Neural Networks (CNNs) or deep-learning techniques.

The sources selected for this review have been guaranteed for their relevance and credibility, selecting from a collection of peer-reviewed research papers, industry publications, and conference proceedings from IEEE and ACM. The review comments on the availability of such sources, highlighting contributions from key journals, conferences, and institutions that have shaped the field. Regarding the papers' publication date, one influential paper by Hinton et al.[5] from the late 2000s written by the father of AI, Geoffrey Hinton is referenced to mark the important milestone in RS, then most of the referenced papers are from recent decade. The evaluation of sources follows a structured approach, first using methodologies to explain the foundational concepts of content-based systems and collaborative filtering. It then employs a chronological approach to illustrate the evolution of RS amidst the AI boom, concluding with a discussion on industry practices, particularly in the case studies of YouTube and Netflix. This organization allows for a coherent narrative that captures the complexity and dynamism of RS.

2 Background

Recommender Systems (RS) are fundamentally derived from Content-Based Systems (CB) or Collaborative Filtering (CF). In this section, I illustrate the impact and dedicated evaluation metrics of modern RS.

2.1 Recommender Systems

RS are specialized tools designed to predict ratings or preferences a user might assign to an item, based on their past behavior and the behavior of others. By analyzing relationships between users and items based on actions and behaviors, RS can make highly accurate predictions without the need for human curation. For instance, Amazon's RS leverages users' past purchasing behavior to suggest relevant products using association rule mining when users checkout (people bought this also bought this). This data-driven methodology has proven not only effective in enhancing user experience but also instrumental in driving significant economic value [12].

One critical aspect of RS is the data source used to understand user preferences. This data can be explicit, like user ratings on products or content, or implicit, inferred from user behaviors such as clicks, purchases, or the amount of time spent on a video. Each data type has its advantages and challenges. For example, explicit feedback can be sparse and subjective, while implicit data like click or purchase data provides a more abundant and objective basis for recommendations but might be misleading in the case of bot fraud and mistake clicks.

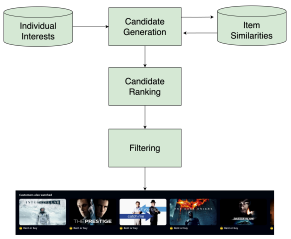


Figure 1: Architecture of a top-N recommender

In practice, RS are often structured as top-N systems, where the goal is to present a finite, optimized list of recommendations to a user. This process typically involves stages like candidate generation, ranking, and filtering, each contributing to the creation of a finely-tuned list of suggestions based on user-aggregated behavior, scores, and machine learning ranking, which is the algorithm published by Amazon [12] in 2003. The complexity and effectiveness of these systems depend significantly on the quality and volume of the underlying data.

2.2 Evaluation Metrics

The methodology for testing RS, especially offline, often employs a train/test split approach familiar to machine learning. In this setup, RS are trained on a subset of user behavior data and then tested on unseen data to predict user ratings. This method, while straightforward, fundamentally tests the system's ability to predict past user ratings, not its core objective – recommending new, interesting items. To refine this process, k-fold cross-validation is sometimes used, providing a more robust assessment by avoiding overfitting to a specific training set.

$$\mathrm{MAE}(f) = \frac{1}{|\mathcal{R}_{\text{test}}|} \sum_{r_{ui} \in \mathcal{R}_{\text{test}}} |f(u,i) - r_{ui}| \tag{1}$$

$$\mathrm{RMSE}(f) = \sqrt{\frac{1}{|\mathcal{R}_{\text{test}}|} \sum_{r_{ui} \in \mathcal{R}_{\text{test}}} (f(u,i) - r_{ui})^2} \tag{2}$$

Among the numeric metrics used for evaluation, Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) are common. MAE in equation 1 assesses the average magnitude of errors in each rating prediction, while RMSE in equation 2, a more stringent metric, assesses the average magnitude of square errors in each rating prediction, aiming to penalize larger errors more severely. However, these metrics focus on rating predictions rather than the system's ability to suggest new items catering to users' preferences.

$$\text{hit rate} = \frac{hits}{users} \tag{3}$$

Given the influential Netflix Prize in 2006 stated by Ricci et al.[1], the winning RS resulted in a 10% improvement in Netflix's own RMSE score (p.82). Although RMSE remains a critical metric, its real-world relevance is debatable. In practice, leave-one-out cross-validation for top-N recommenders more often utilizes hit rate as an evaluation metric, measuring the accuracy of the top-N lists of individual users instead of each individual rating, which is more focused on user interactions. In this approach, one item in each user's data is held out to test the model's predictive capability, and if the RS predicts this item within the top-N suggestions, it counts as a hit. The hit rate is then determined by the proportion of hits across all users using equation 3. It offers a nuanced understanding of RS's effectiveness, prioritizing user satisfaction over mere predictive accuracy. Beyond accuracy, other important metrics include coverage, diversity, novelty, churn, and responsiveness regarding the systems.

Despite the plethora of offline metrics, real-world testing through online A/B tests remains paramount. These tests observe how actual users interact with recommendations, thereby directly measuring the impact on user behavior and business outcomes. They help in making practical and realistic decisions about the system's complexity and effectiveness.

2.3 Content-Based Systems

Starting with the simplest approach, content-based systems, we consider recommending items based solely on their attributes rather than relying on aggregate user behavior data. For instance, in the context of movie recommendations, this could involve suggesting films from the same genre or release year as those previously enjoyed by a user. The incorporation of detailed attributes, such as director, actor information, or critic review scores, can further refine these recommendations. The MovieLens dataset [13], often used in such systems, provides genre classifications and release years, forming a foundation for CB systems.

$$\text{CosSim}(x, y) = \frac{\sum_i x_i y_i}{\sqrt{\sum_i x_i^2}\sqrt{\sum_i y_i^2}} \tag{4}$$

Mathematically, the similarity between items (e.g., movies) based on their attributes can be measured using metrics like cosine similarity by equation 4 from Ricci et al.[1] (p.124). This approach takes the summation of the genre of products between X (movie 1) and Y (movie 2), and divides it by the products of square roots of summations of the squared value of each individual movie's genre, ending up with a cosine similarity score. The cosine similarity metric is versatile and can be applied across various recommendation models, not just CB systems.

Research by Deldjoo et al.[2] in the field has explored more novel approaches, utilizing mise en scène data in movie recommendations. This involves analyzing properties of movie scenes, such as shot length, color variance, and scene lighting, to recommend films based not just on genre

or release year but on the mood and pacing of the films themselves. While such bleeding-edge approaches can increase the diversity of recommendations, their impact on accuracy and user satisfaction can vary and often requires further real-world testing.

2.4 Neighbourhood-based Collaborative Filtering

Neighborhood-based collaborative filtering is a cornerstone of modern RS, leveraging the behavior of similar users or items to predict what a specific user might like. This technique relies on the fundamental concept of "neighborhoods", which refers to groups of users with similar preferences or items that share common characteristics [1]. To effectively implement neighborhood-based CF, each attribute of users or items is considered a dimension in a multi-dimensional space, measured by the cosine similarity between different entities in this space.

In user-based collaborative filtering, the process involves identifying users similar to the target user and recommending items liked by these similar users. This method constructs a user-to-user similarity matrix and generates recommendations based on the ratings of the most similar users [1]. Conversely, item-based collaborative filtering focuses on identifying items similar to those a user likes, presenting a few advantages: items tend to be more stable than user preferences, there are usually fewer items than users, making the computation more efficient, and it allows for immediate recommendations to new users [1]. Recent advancements in the field include research by He et al.[3] on translation-based recommendations, which use vectors to model users moving from one item to another in a multidimensional space. This approach has shown promising results in predicting sequences of events, such as which movie a user is likely to watch next, based on their viewing history.

CF has proven effective, particularly in large-scale applications. Despite criticisms of limited scalability and sensitivity to noisy and sparse data, item-based CF has notably reduced complexity, making it feasible to compute similarity matrices for extensive product catalogs on single machines. However, the challenges of data sparsity remain as this research proposed. Most users have only seen a fraction of the available items, leading to a predominantly empty matrix of user-item interactions. This sparsity hinders the effectiveness of CF, particularly in systems with a smaller user base. Companies with large user bases, like Amazon and Netflix, can leverage their extensive data to overcome these limitations.

3 Conventional Methods in Recommender Systems

This section highlights traditional methods in industry practice.

3.1 Matrix Factorization Models

To advance beyond the limitations of CF, model-based methods have been explored. These methods apply machine learning techniques to derive predictions from user ratings, moving away from similarity comparisons toward training predictive models. Principal component analysis (PCA) exemplifies a matrix factorization technique by reducing the dimensionality of data and identifying principal components [1] (p.248). The PCA approach has been used to distill user ratings into lower dimensions that capture the most variance within the data, often aligning with recognizable attributes such as movie genres. This process transforms the sparse matrix of user-item ratings into factors of smaller matrices describing general user and item categories.

$$R = U\Sigma M^T \tag{5}$$

Singular Value Decomposition (SVD) is based on PCA computation and has been widely recognized for its accuracy. From equation 5, SVD efficiently computes the user matrix U, scaling diagonal matrix Σ, and transposed movie matrix M^T all at once for reconstructing rating matrix R to predict unknown ratings. However, when dealing with missing data, SVD-inspired algorithms like SVD++ and stochastic gradient descent (SGD) or alternating least squares (ALS) come into play, optimizing the factor matrices to minimize the error in known ratings [1].

3.2 Restricted Boltzmann Machines

The early-type neural network model for recommender systems, Restricted Boltzmann Machines (RBMs), has been a mainstay since 2007, demonstrating comparable performance to SVD. The integration of RBMs with matrix factorization methods (SVD), as evidenced in the Netflix Prize, suggested that a combined approach could yield superior results, which Netflix claimed part of its RS still using this approach a few years ago [1]. RBMs rely on a two-layered structure, encompassing a visible layer to intake user ratings and a hidden layer where the training through forward and backward passes takes place [5].

4 Emerging Research in Deep-learning Recommender Systems

This section analyses cutting-edge research and emerging techniques in AI that have the potential to revolutionize RS but might not yet make it into the mainstream.

4.1 Autoencoders

AutoRec offers a modern approach to RS, utilizing a simple architecture with an input layer for the individual ratings of each user, a hidden layer, and an output layer for predicted ratings. A matrix of weights and bias terms between the layers with sigmoid activation output functions is maintained across the hidden and output layers, involving the encoded and decoded stages for separate sets of weights stated in Sedhain et al.[6]. Despite the fundamental difficulty of distinguishing between missing ratings and rating of zero remains a significant challenge, a transformer-based deeper network published by Keras [14] can help mitigate this issue by leveraging the power of attention mechanisms to capture complex patterns and dependencies in the data, these networks have shown marginal improvements over RBMs.

4.2 Clickstream Recommendations with Recurrent Neural Networks

Recurrent Neural Networks (RNNs), with their prowess in analyzing patterns in sequences of data, lead to session-based recommendation systems. These systems are particularly applicable to e-commerce platforms and streaming services, where user sessions provide a sequence of previous interactions to inform the next recommendations through real-time clickstreams. The GRU4Rec model, utilizing Gated Recurrent Units (GRUs) which learn when and by how much to update the hidden states within each unit, represents a customized RNN framework designed to optimize session-based recommendations in Hidasi et al.[7], featuring session-parallel mini-batches, sampling outputs for instant recommendations, and new ranking loss function for

RS application. This highlights the potential of deep learning in this area. While RNNs using session click order showed slight improvements over traditional techniques in [7], complex deployments, adoptions, low latency, and maintainability in a real-world setting are crucial for RS, like instant web page rendering, and fine-tuning. Hence, a simpler and effective system is often preferable unless the new technology proves to be significantly superior.

4.3 Neural Collaborative Filtering

Neural Collaborative Filtering (NCF) is a compelling algorithm within the RS domain recently, merging the strengths of matrix factorization's robust history and the nuanced capabilities of NNs stated in He et al.[8]. Matrix Factorization, despite its longevity and relative simplicity, is unable to capture non-linear relationships, which is a gap that NNs can fill. NCF hybrid model harnesses the predictive powers of both by running user and item data through a Generalized Matrix Factorization model (GMF) and a Multi-Layer Perceptron (MLP) in parallel.

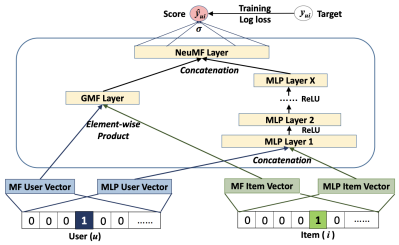


Figure 2: Neural Matrix Factorization model [8]

The process, shown in Figure 2, begins with a one-hot encoding of a user and an item, branching into separate embeddings that feed into the GMF and MLP pathways. The GMF utilizes an element-wise product to merge user and item embeddings, while the MLP concatenates these embeddings to provide a deep learning foundation. Following the training through these parallel structures, the outcomes are merged in the neural matrix factorization layer with a sigmoid activation output function, producing rating predictions given user-item rating pairs. The concept illustrates the MLP's ability to delve into the depth of the model, facilitated by functions like ReLU, and the GMF's focus on more direct interactions between users and items.

4.4 Deep Factorization Machines

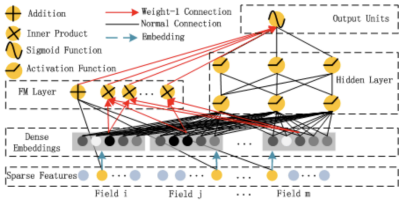


Figure 3: Wide & deep architecture of DeepFM [9]

Deep factorization machines (DeepFM) in Guo et al.[9] merge factorization machines' generalizability with the analytical depth of NNs. In figure 3, DeepFM operates by feeding raw feature

data into both a wide factorization machine and a deep neural network simultaneously, seeking to capture both low and high-order feature interactions respectively, more than just the interactions between users and items captured by the NCF. The outputs are combined to predict user actions, such as clicks or purchases. DeepFM's ensemble approach is theoretically sound and more powerful because it is technically the enhanced version of the hybrid model from the winning Netflix price with a deeper neural network (originally RBMs) and a more versatile Factorization method (originally SVD). It therefore could be a potential and successful hybrid model in a scaled-up production world.

4.5 Generative Adversarial Networks

Generative Adversarial Networks (GANs), another area of intense research focus, offer a distinct methodology for RS. Recalling the fundamental GAN architecture, where a generator and discriminator work in opposition to produce realistic outputs, the application in RS substitutes images with ratings. In Bharadhwaj et al.[10], the generator aims to fabricate a full matrix of user-item ratings, while the discriminator discerns between these and actual observed ratings. Unlike traditional GANs, convolutional layers are replaced with RNNs with GRUs due to the temporal nature of rating data, and the model inputs historical ratings instead of random noise to generate future predictions specific to particular users.

GANs have shown efficacy in handling sparse data, a common challenge in recommendation contexts. They can adapt to both dense and sparse datasets, as evidenced by their application to subsets of Netflix data. However, their computational intensity cannot be understated, with the generation of user-item pairs for millions of users presenting a significant challenge.

5 Case Studies: YouTube and Netflix

This section analyses the innovative RS and the real-world challenges of Google and Netflix.

5.1 YouTube

YouTube's recommender system confronts several unique challenges stated in Google [11]. Firstly, the sheer scale of data, with a user base exceeding a billion and daily video views reaching billions, necessitates an infrastructure capable of horizontal scaling and rapid content delivery. Freshness is another main challenge, with hundreds of hours of content uploaded every minute, requiring the recommender to swiftly incorporate new content while maintaining a balance with established videos. Additionally, the system deals with the noise inherent in implicit ratings, such as clicks and search queries, which can be unreliable indicators of user preferences.

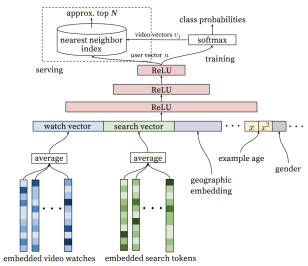


Figure 4: YouTube's deep candidate generation model architecture [11]

To address these challenges, YouTube has adopted a deep learning model that maps the averaged sparse behavioral data into a denser, fixed-length format suitable for neural network input through a self-learned embedding layer, which can solve the data sparsity issue, scalability, and high computation power. Despite having explicit ratings available, YouTube disregards this sparse data in favor of implicit signals, which offer a more abundant data source. The deep learning architecture then concatenates these signals with user demographic data to generate the expected watch time of each top-N candidate using "learning to rank", prioritizing actual user engagement over mere clicks for more personalized systems. The three hidden layers, the first has a width of 1,024 units, followed by 512, and then 256 at the top, produce a good balance of accuracy while staying within their budget for computing hardware [11].

YouTube exemplifies the complex interplay between user behavior, content freshness, and the need for scalability in recommender systems. It highlights the significance of deep learning in managing vast datasets and the delicate balance between recommendation relevance and user engagement optimization.

5.2 Netflix

Netflix has selectively disclosed its recommendation system's architecture and strategies mentioned in the case study of Ricci et al.[1] (p.399). Contrary to YouTube's deep learning-centric approach, Netflix's stance on algorithmic diversity implies an experimental and inclusive methodology, likely encompassing a suite of techniques from linear regression to k-means clustering, suggesting an exploratory disposition toward algorithm application. One of the critical lessons highlighted by Netflix Prize is the shift from a focus on accurate rating predictions to a more nuanced approach emphasizing rankings and whole-page optimization.

Netflix's RS transforms all user interactions into potential data for improving user experience. The user interface, segmented into various category rows, functions as distinct recommender engines, each customized for different content types or genres. This complex structure necessitates careful management of non-redundancy and the optimization of recommendation presentation order, known as whole-page optimization [1]. Both platforms share a commonality in their method of learning to rank, where machine learning is applied to the sequence of top-N recommendations, using additional context beyond rating predictions to refine the final output. This process includes an intentional injection of popular content into the recommendations to foster user trust and balance the individualized suggestions with broadly appealing selections.

Netflix harnesses a combination of diverse algorithms, personalized content discovery, and strategic content placement to create a dynamic and engaging recommendation environment, underscoring the effectiveness and industry trend of evolving recommender systems to prioritize meaningful user engagement and content discovery over traditional predictive accuracy metrics.

5.3 Real-world Challenges and Considerations

Several existing challenges in RS must be addressed. The "cold start problem" poses a significant issue, particularly when tackling new users or introducing fresh catalog items lacking historical data. For new users, leveraging implicit behavioral signals and employing techniques, like geo-IP analysis and top-seller recommendations, can provide initial data points for preliminary recommendations. New items may benefit from a content-based approach, using attributes like description, category, or mapping attributes to latent features in Deborah et al.[15]. Ensuring the exclusion of sensitive content through stoplists is crucial to maintaining social norms

and legal requirements. Temporal dynamics, including seasonal variations and recency of interactions, significantly influence recommendation accuracy, necessitating the weighting of recent interactions or incorporating time as a feature. Balancing business profitability with user interests is another key aspect, where value-aware recommendations align with business objectives but must not compromise user's trust and engagement. Lastly, while deep learning models offer promising results, practical considerations such as interpretability, transparency, hardware demands, and low latency must be weighed against their benefits, and simpler models like item-based K-nearest neighbors RS may offer competitive solutions in production RS while ensuring the system design will not introduce and deepen the representational and allocative bias.

6 Summary & Conclusion & Future Research

The literature review provided offers a comprehensive analysis of the evolution of RS in the age of deep learning and AI. The key findings highlight a significant transformation from traditional content and collaborative filtering methods to more sophisticated deep-learning models. The significance of these advancements lies in their capacity to handle the scale, noise, and sparsity of data in modern applications, greatly enhancing personalization and user engagement.

The review addresses gaps in existing knowledge by showcasing how contemporary RS, particularly those using deep learning, can effectively manage large datasets, and improve upon the limitations of traditional models, such as scalability and data sparsity. It contributes new insights into the integration of AI in RS, emphasizing the importance of practical and ethical considerations in various industries.

Regarding the proposed future research directions, it is recommended to delve deeper into the integration of multimodal data sources, such as combining visual, audio, and textual data, to create more nuanced and personalized user experiences. This could involve developing advanced 3D CNNs in Tuan et al.[16] for analyzing clickstream data and audio waveforms in Chang et al.[17] for music recommendations, reflecting the latest trends in edge AI products like 'Spotify DJ' music assistant released by Spotify [18]. Open-source collaborative ecosystem should be emphasized in the AI field by the tech giants which have the resources and expensive infrastructures to build these large models, allowing individual developers and startups to leverage the open-sourced efforts that could make AI development more accessible, fair, and well-rounded. For instance, Meta open-source Llama 2 allowed more research in the university and public communities; Google on YouTube's deep-learning architecture on RS propelled the good competition within the industry; Amazon's open-source system, DSSTNE, stated in article [19] exemplifies the application of neural networks to RS, boasting efficiency in handling sparse data and scalability, and the accessibility of RS technology in the business world.

The literature review effectively demonstrates that the proposed research objectives have been accomplished by providing a thorough understanding of the current RS and outlining a clear path forward. The work done contextualizes the significance of RS within the field of all state-of-the-art AI research headings, highlighting their potential impact on user engagement, ethical AI development, and the democratization of AI technologies.

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